INCHEON RoboCup 2025 | U.S.A | FATEN ROBOTICS | Rescue Line Team L11

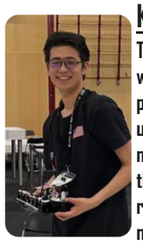
Team Information:

We are Team FATEN and we will be representing The United States of America in Robocup Junior Rescue Line 2026! We are a global team, combining expertise and insights from all over the world, and from multiple previous teams. Therefore we have a long and wide spanning track record in all types of robotics. We placed 16'th, and won the comunity award at the 2024 world championship. We have also seccured three victories at american regionals.

Asia Pacific 2021

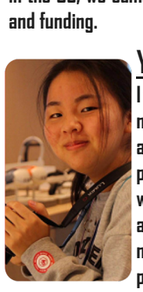
Asia Pacific 2023

World Championship 2024



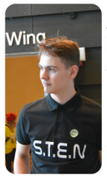
Kent Nakai - Head developer

This is my tenth year in RoboCup. When I started at nine, I only wanted to make a robot move forward. Since then, competition has pushed me to build robots that adapt to their environment and to understand what it means to build something as a team. After moving to the US four years ago, I built teams from scratch at three different schools and won U.S. Nationals three years in a row, though not without many failures. Last year, our robot could not even drive straight through an entire competition. Through all of this, what grew the most was not only technical skill, but my understanding of other people and myself. This year brings all of that together. With Hampus, whom I met in the Netherlands, friends I used to compete with in Japan, and friends I made in the US, we came together to build one robot that balances hardware, software, and funding.



Yuna Tamura - Hardware and Management

I focus on hardware development, while also handling most of the managerial work, as well as taking responsibility for sponsorship and public relations. Hardware support is one of the roles I'm most proud of — I genuinely enjoy building, problem-solving, and working hands-on with the team. Beyond the technical work, I care a lot about supporting the team in every way I can. Sometimes that means coordinating logistics, designing our team apparel, or photographing our activities so we can share our progress with others. I've competed in RoboCupJunior in Japan, and in 2021 I placed 2nd in Rescue Line Primary at RCAP Nagoya. That experience shaped how I approach teamwork today: combining technical development with the behind-the-scenes work that keeps everything running smoothly.



Hampus Mollgård - Mentor and advisor

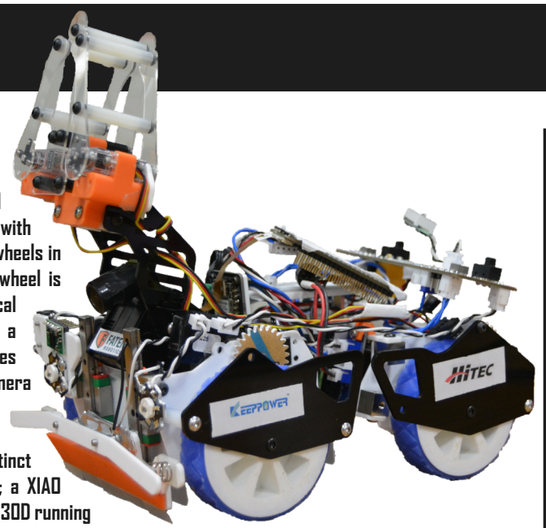
With a long background as team leader of a Swedish robotics team, I first met Kent, and Team FATEN as a competing team in Eindhoven 2024. Since then, we have stayed in contact and developed this international collaboration. I reprecented Sweden both in the 2024 world championships finnishing 19'th, and the 2025 world championship finnishing 8'th. We also won the 2025 comunityaward.

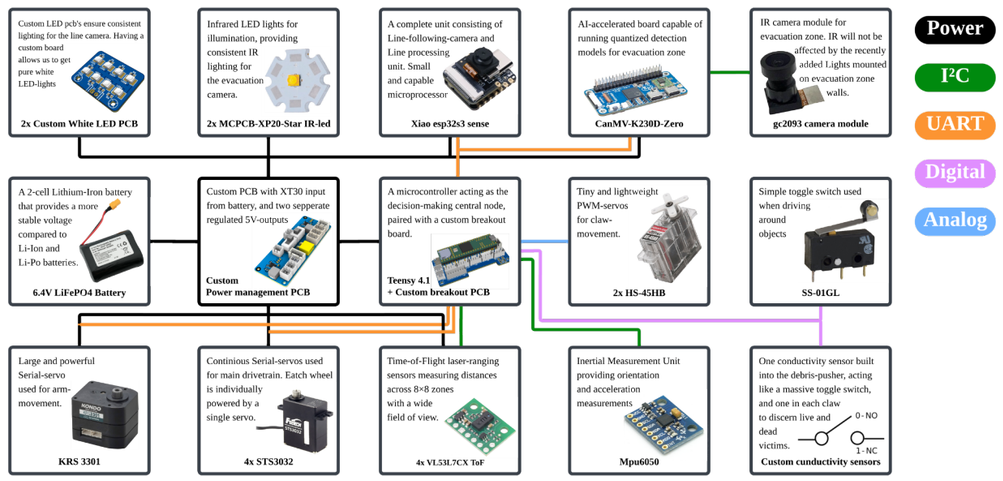
Joshua Enelamah - Advisor

I competed alongside Kent last season, and I saw first-hand how many ideas this team wanted to fit onto a single robot. That ambition exited me to want to keep working with him and FATEN. This year I continue that collaboration from last year mainly on Electronics and Software Ideas, so the rest of the team can push those ideas as far as the hardware will allow.

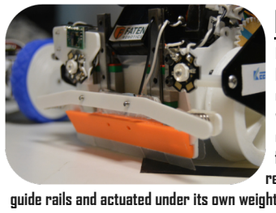
Hardware:

The 2026 FATEN robot is built around one idea: repeatable control in unpredictable terrain. It uses a compact 140×180mm chassis with a flexible enhanced-acrylic suspension plate that keeps all four wheels in contact with the ground even when tile height changes. Each wheel is independently driven by an STS3032 serial servo. The mechanical structure includes a lightweight dual-grip rescue arm and a front-mounted, linearly guided debris-removal unit. The robot uses two separate cameras: a downward-facing line-following camera mounted under the front unit, and a front-facing 850nm IR camera for victim detection. The electronics are distributed across custom-designed PCBs and three controllers with distinct roles: a Teensy4.1 for motor control, sensing, and state logic; a XIAO ESP32-S3 with an OV2640 for line and feature detection; and a K230D running a YOLOv8n model for evacuation-zone object detection.



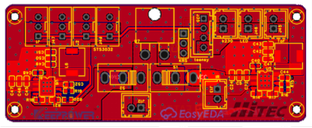


Innovative solutions:



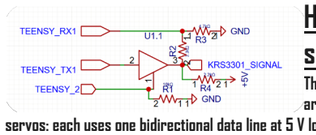
Debris-remover

Our robot features a frontmounted linearly guided debris-pusher capable of following uneven terrain while still blocking debries from entering the cone around the line-following-camera. This debris-remover is based on LWL7B linear guide rails and actuated under its own weight and gravity.



Custom Power PCB

We fix last year's noise in the layout itself. The motor power and the logic power sit on their own copper, and both grounds come back to one star point by the battery input, so the motor's current spikes do not move the logic ground. The bulk and decoupling caps sit right at the regulator and at each chip's power pin, so the switching current stays in a small loop and does not radiate into the sensor lines. For heat, the regulator pad ties into a large GND copper pour through a grid of thermal vias, so the ground plane spreads the heat out instead of it building on the chip. With this the supply stays stable and the controller does the same thing every run.



Half-duplex servo driver

The KRS3301 and STS3032 are half- duplex serial servos: each uses one bidirectional data line at 5 V logic, while the Teensy has separate RX and TX pins at 3.3 V. We bridge both gaps with a 74HCT126 tristate buffer. When sending, the HCT buffer reads the Teensy's 3.3 V TX as a valid HIGH and drives the line at a clean 5 V, and its tristate enable releases the line so the servo can reply without contention. When receiving, the servo's 5 V output is dropped to 3.3 V through a voltage divider before it reaches the Teensy. One UART now drives both servos cleanly, level-shifted in each direction.

IR Camera - Victim detection

The 2026 rules introduced perpendicular LED lights in the evacuation zone, so we upgraded to an IR-based camera. We built it ourselves by removing the RGB filter and adding an 850nm bandpass lens. Since LEDs emit almost no IR, the field lighting has minimal impact on the image, even at competition brightness. In IR, silver reflects strongly while black absorbs, making victim detection much cleaner. We first tested 940nm, but it interfered with our ToF sensors. Switching to 850nm and adding a matching bandpass filter removed that issue. With this setup, the AI model produced promising results.

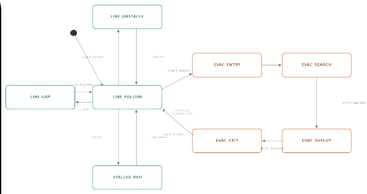
Software:

Software overview

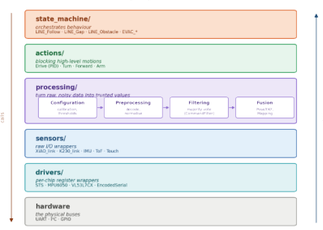
The software runs on three boards in two languages. The XIAO ESP32-S3 runs the camera vision, the K230D runs the AI model, and the Teensy 4.1 is the conductor that decides what the robot does. The Teensy and the XIAO are both Arduino (C/C++); the K230D is MicroPython. The two co-processors do the heavy seeing, so the Teensy is left to think and act. Beacuse of this simplified arcitecture, with outsourced tasks, and a desition-making state machine, our main loop becomes really simple. The main loop is only consists of our state machine and its flowchart, explained below.

State machine

The robot's behaviour is controlled by a state machine. It spends most of the run in its main line-following state, and switches to other states only when certain events occur — for example when it detects an obstacle, loses the line, or enters the evacuation zone. Simple actions like U-turns and green-tile turns are handled directly inside the line-following state, since they don't require changing to a different mode. Each state is a small, self-contained block of code. This makes the system easy to maintain: adding a new behaviour or fixing an existing one doesn't affect the others.



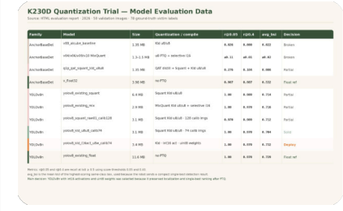
Layered structure



The Teensy code is organised in layers, where each layer mainly talks to the one below it. At the top, the state machine decides what the robot should do. Below that, the actions layer handles the blocking motions such as PID-based driving, turning, and the arm. The processing layer is where raw sensor data becomes something the robot can trust. It works in four steps. Configuration applies calibration and thresholds so the same code adapts to different arena lighting. Preprocessing decodes and normalises the raw bytes from the XIAO and K230D. Filtering removes noise — for example, a command is only accepted after a 15-frame majority vote. Finally, fusion combines the cleaned values into a pose estimate and a map that the state machine can use.

AI integration

The main evaluation focus was the evacuation-zone AI model, because slope driving had already been tested through the mechanical layout, suspension, and terrain-biased PID controller. For AI, the difficult part was not only training a detector, but proving that the model still worked after preprocessing, quantization, and conversion to a K230D kmodel. Our evaluation pipeline started from raw RGB images collected on the robot. We then generated several dataset variants in Python, including raw RGB, shadow-separated images, and CLAHE / histogram-equalized images. Each variant was used to test whether preprocessing improved victim contrast before model training.After that, we trained and compared several detector families, including



Thanks to everyone who has supported us throughout this journey — teachers, friends, parents, previous teammembers, competitors, sponsors, and the RoboCup Federation!

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